

Data Science

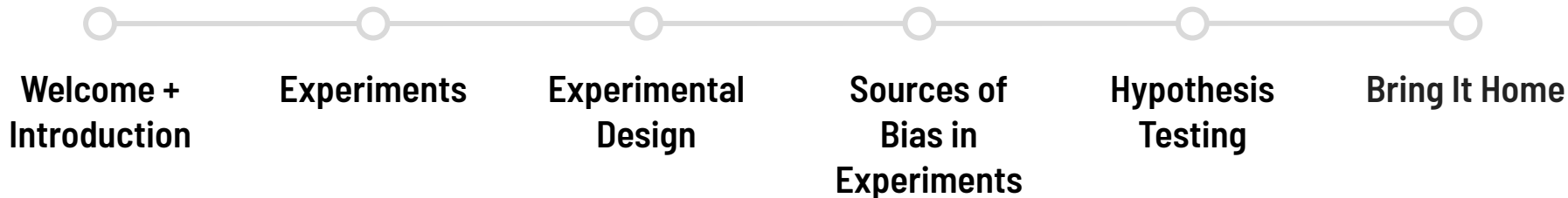
Experimental Design

Exploratory Data Analysis



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Explain the main reasons to conduct experiments.

2

Understand the framework for conducting an experiment.

3

Define key components of an experiment including the outcome, sources of variation, and the necessary data.

4

Explain how experimental design relates to A/B tests.

5

Define the key steps and terms in hypothesis testing.

What are experiments?

Experiments

You're an analyst for BIG Sodas, a producer of soft drinks.

They want to bring out a sugar-free version of their most popular lemonade drink, BIG Lemon.

They have two formulas to test: BIG Lemon Zero and Diet BIG Lemon.



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They want to bring out a sugar-free version of their most popular lemonade drink, BIG Lemon.

They have two formulas to test: BIG Lemon Zero and Diet BIG Lemon.



- They want you to **help design an experiment to decide which of the two versions to produce.**
- The basic idea of the experiment is to do blind taste tests to see which flavour people prefer.

What is an

experiment?

What are experiments for?

Determine causes

To determine the principal causes in a measured response.

Finding extremes

To find the conditions that give rise to a maximum or minimum response.

Comparing effects

To compare the response achieved at different settings of controllable variables.

Predict the future

To obtain a mathematical model in order to predict future responses.

What are experiments for?

Determine causes

To determine the principal causes in a measured response.

Comparing effects

To compare the response achieved at different settings of controllable variables.

These are what we might call
“inferential statistics”

What are experiments for?

These are what we might call
“machine learning”

Finding extremes

To find the conditions that give rise to a maximum or minimum response.

Predict the future

To obtain a mathematical model in order to predict future responses.

Planning the Experiment

- Why conduct an experiment?
- Which of the reasons to conduct an experiment apply to the soda experiment?
- How do they apply?

- ☐ To **determine the principal causes** in a measured response.
- ☐ To find conditions that give a **maximum or minimum response**.
- ☐ To **compare the response achieved at different settings** of controllable variables.
- ☐ To obtain a mathematical model to **predict future responses**.

Experimental design

How to Design an Experiment

1

Objectives

Define the objectives.

2

Variation

Identify all possible sources of variation.

3

Assign

Choose a rule for assigning units to treatments (e.g. which patients get drugs vs. placebos).

4

Measurement

Decide on measurement to be made and analysis procedure.

1. Define the objectives

- You should make a precise list of the **questions you'll want to be able to answer** after the experiment. Feel free to use the word **cause** here!
- Only list the **essential questions** you'd like to answer.
 - *Too many questions*, and you unnecessarily complicate the experiment and increase chances of mistakes.
 - *Too few questions*, and the ones you omitted may be unanswerable from your experimental design.

Optional reading to dive into causality: ***Causal Inference in Statistics: A Primer (Pearl)***

Defining the Objectives

What are the questions we want to answer?

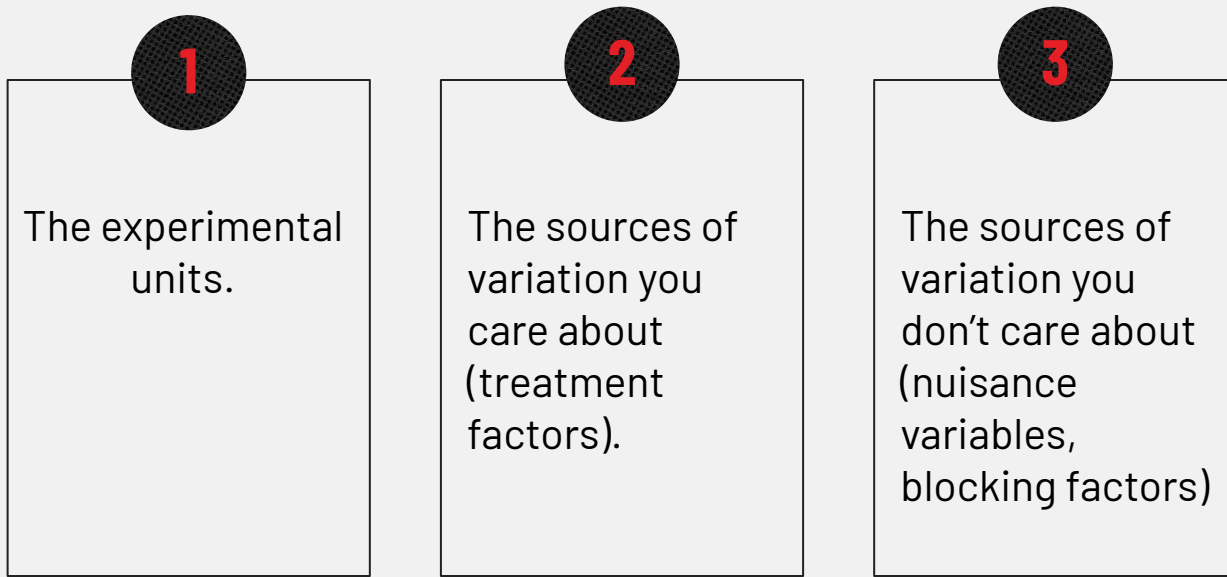


Reminder of the case study

BIG Sodas want to design an experiment to decide which of the two versions of their lemonade to produce: BIG Lemon Zero or Diet BIG Lemon

2. Identify sources of variation

Let's divide all sources of variation into three categories:



What are “experimental units”?

The experimental unit of an experiment is the basic “thing” you’re experimenting on.

If you’re having trouble identifying your experimental unit, try to picture what you want your data to look like at the end of your experiment.

Your experimental unit is probably what each row represents.

What is our experimental unit?

What are “treatment factors”?

A treatment factor (or just **treatment** for short) is **a source of variation that you wish to manipulate** manually to measure how it affects the response. A treatment may have two or more levels.

In many experiments, you might wish to have a **control** treatment level (the experimental unit receives nothing) and/or a placebo treatment level (the experimental unit receives a fake treatment).

What is our treatment?

Do we have a control?

What are “blocks”?

A blocking factor (or **block**, for short) is **a source of variation that affects your response** in a way that you might consider it a *nuisance* variable.

It's not a variable that you can manipulate as part of your experiment, yet it may be interesting to measure the differences between blocks.

Examples might include patient demographics (age, gender etc.) in a drug trial.

What are “blocks”?

Treatment factors are things **you administer** as part of the experiment. You may care about how treatments affect the response in different blocks.

For example, will Medicine A or Medicine B improve a patient’s health more? This will differ by patient age, so you separate patients into three categories: “Below 18”, “18-54”, and “55 and up”.

The **medicine is the treatment factor**, while the **age category is a blocking factor**.

What are some blocks in the soda experiment?

The 2008 Obama Campaign

The team behind the campaign wanted to see which combination of **main image** and **buttons** resulted in the **greatest number of email addresses given**.



The 2008 Obama Campaign

Experimental units: people viewing the campaign

Treatment factors: combinations of hero image + buttons

Experimental design: show random combination of image + buttons to people, count how many people give their email per combination.



The 2008 Obama Campaign

The campaign tested **six** different images and **four** different buttons to see which combination resulted in the **greatest number of email addresses** given.

There were two treatment factors with 6 and 4 levels, respectively.

There were thus **24 total treatment combinations**.



The 2008 Obama Campaign

The experiment was extremely successful. By using the best combination of image and button, they received **40% more email addresses**, from 7.1 million to 10 million.

The average donor gave \$21. That's a total **increase of \$60 million!**

Read more about this [here](#)!



The 2008 Obama Campaign

The result!



The image is a screenshot of the Obama 2008 campaign website. At the top, there is a blue header with the Obama 'O' logo and the text 'OBAMA'08'. Below this, the word 'CHANGE' is prominently displayed in large blue letters, with the tagline 'WE CAN BELIEVE IN' underneath it. The central part of the page features a black and white photograph of Barack and Michelle Obama with their two children. Below the photo, there is a sign-up form with the text 'JOIN THE MOVEMENT' on the left. The form includes two input fields: 'Email Address' and 'Zip Code'. To the right of these fields is a red button with the text 'LEARN MORE'. At the bottom of the page, there is a blue footer bar. On the left, it says 'PAID FOR BY OBAMA FOR AMERICA'. In the center, there is a small circular logo. On the right, it says 'CONTINUE to WEBSITE'.

3. Choosing a rule to assign units to treatments

The most important rule is **randomness**.

An experiment in which units are randomly assigned to treatments is called a **Completely Randomized Design (CRD)**. *(If there are blocks involved, you could also call it a Completely Randomized Block Design (CRBD).)*

Randomness is the **only** way to ensure no unaccounted for factors are affecting your design.

3. Choosing a rule to assign units to treatments

Sometimes you can't randomly assign something. For example, if you want to see the effects of smoking on lung health, it would be unethical to randomly assign individuals to various smoking levels.

If your treatment (e.g. smoking) can't be randomly assigned, you can only look at historical data. This is no longer an experiment, but an **observational study**.

Observational studies are fine and useful, but **they cannot show causation**.

3. Choosing a rule to assign units to treatments

Another choice we might have to make is between:

WITHIN-SUBJECT DESIGN	BETWEEN SUBJECT DESIGN
Every subject experiences every treatment. Variation is measured for each subject. Each subject is their own control.	Subjects are assigned into treatment and control groups. Variation is measured between groups. An entire group acts as a control.

3. Choosing a rule to assign units to treatments

In the soda experiment:

WITHIN-SUBJECT DESIGN	BETWEEN SUBJECT DESIGN
Everyone tries both sodas. Everyone rates both sodas. We focus on the difference between ratings.	Every participant only tries one soda. Everyone rates only one soda. We focus on the difference in group average ratings.

Choosing a Study Design

In groups, discuss:

- ☐ The **pros and cons** of within-subject and between-subject design for the soda experiment specifically
- ☐ What are the implications of choosing one or the other? Think about:
 - Costs/logistics
 - Ability to draw conclusions
 - Real-world applicability

Within-subject design:

- ☐ Everyone tries both sodas.
- ☐ Everyone rates both sodas.
- ☐ We focus on the **difference between ratings**.

Between-subject design:

- ☐ Everyone only tries one soda.
- ☐ Everyone rates only one soda.
- ☐ We focus on the **difference in group average ratings**.

4. Decide on measurements to be made

- ❑ How will **measurements** be made? What are you measuring to determine answers to your questions?
- ❑ How many treatment levels are there?
- ❑ Are there any blocks?
- ❑ What pitfalls can you run into?
- ❑ Do you need **repeated measurements**?

What measurements will we take in the soda experiment?

Recap: Designing an Experiment

Steps	Soda case study example
1. Define the objectives .	Which type of low sugar soda do people prefer more? Determine by conducting and analyzing taste tests
2. Identify sources of variation : - Experimental units - Treatment factors - Blocking factors	• Experimental units are the participants. • The treatment factor is the type of soda. • Blocking factors include demographics (age, gender, etc.), existing preferences, or time/day of taste tests
3. Choose a rule to assign units to treatments.	Participants randomly allocated Need to decide between within-subject or between-subject design
4. Decide on measurements .	Measurement will be a rating (e.g. “how tasty is this soda out of 10”)

Sources of Bias in experiments

How do experiments
go wrong?

Recap: Designing an Experiment

Steps

1. Define the **objectives**.

2. Identify sources of **variation**:

- Experimental units
- Treatment factors
- Blocking factors

3. Choose a rule to **assign** units to treatments.

4. Decide on **measurements**.

How can each step go wrong?

1. Define the objectives

Sources of bias/problems:

- ❑ The questions **cannot be answered** with the gathered data.
- ❑ The objectives are **misaligned** with the business problem.
- ❑ There are **too few** or **too many** objectives.
- ❑ The objectives do not lend themselves to a randomized experiment.

2. Identify sources of variation

Sources of bias/problems:

- The **treatment is wrong**/inappropriate/doesn't cause variation.
- Blocking factors are omitted, resulting in **unaccounted for variation**.
- Treatment blocks **vary in size** (e.g. many more people get placebos than medicine).

3. Choosing a rule to assign units to treatments

Sources of bias/problems:

- Participants **aren't random** (e.g. selection bias).
- Treatment **blocks are created without other blocking factors** in mind (e.g. separating people by age but not by gender - only a problem if both are relevant!).
- **Wrong study design**, i.e. using within-subject design instead of between-subject design (or vice versa).
- **Wrong control** selected (or none when there should be).

Selection Bias

Selection bias is where participants are selected in a non-random way.

This could mean:

- ❑ Running an experiment for a subset of **people that aren't representative** of the entire population.
- ❑ **Self-selection bias** where participants ask to take part in the experiment.

4. Decide on measurements to be made

Sources of bias/problems:

- ❑ Measurement is **inappropriate** (e.g. a binary “did this medicine work?” instead of a numeric rating).
- ❑ Measurement is **not detailed enough** (e.g. only measuring a change in heart rate but not other factors).

Recap: Bias in Experiments

Steps	Sources of bias/problems
1. Define the objectives .	Objectives inappropriate or misaligned to business problems
2. Identify sources of variation : - Experimental units - Treatment factors - Blocking factors	Wrong treatment selected Inappropriate or unaccounted-for blocks
3. Choose a rule to assign units to treatments.	No control group when there should be Non-random assignment of units to treatments Wrong study design
4. Decide on measurements .	Inappropriate measurement Measurements too simple or too few

Identifying Sources of Bias

In groups, discuss:

- What sources of bias can you identify for each step when designing the soda experiment?

Steps

1. Define the **objectives**.

2. Identify sources of **variation**:

- Experimental units
- Treatment factors
- Blocking factors

3. Choose a rule to **assign** units to treatments.

4. Decide on **measurements**.

Hypothesis Testing

Definition of Hypothesis Testing

Hypothesis testing is the **scientific act of using statistics** to determine if your hypothesis is likely true within probabilistic reason.

You begin by specifying two hypotheses:

The null hypothesis

The “conventional wisdom” you seek to disprove, or the “base” belief.

The alternative hypothesis

The exciting finding you want to prove.

Key Terms

Hypothesis

A **hypothesis** is a measurable statement that can be verified .

Null hypothesis

The **null** hypothesis is the state we are looking for evidence **against**.

Alternative hypothesis

We are looking for supporting evidence for an **alternative** hypothesis.

Statistical test

A **statistical test** let's us quantify the likelihood of seeing our experimental data in a world where the null hypothesis is true .

Test statistic

A **test statistic** is the value we use to determine whether to reject the null hypothesis.

In hypothesis testing, we
are

**not “proving”
anything**

Hypothesis Testing

1

Design the experiment and choose null and alternative hypotheses.

2

Conduct the experiment, gather experimental data.

3

Use an appropriate statistical test to obtain the result of the experiment.

4

Based on the test statistic, either:

- **Reject** the null
- **Fail to reject** the null

We never “prove” the alternative hypotheses!

What hypothesis testing **is** and **isn't**

Rejecting the null

Rejecting the null hypothesis means:

"The likelihood of observing this data if the null hypothesis is true is **too low**."

Rejecting the null

This does not mean we prove the alternative hypothesis, **only that we don't believe the null**.

Failing to reject the null

Failing to reject the null hypothesis means:

"There is **not enough evidence** to suggest this data could not be obtained if the null hypothesis is true."

Failing to reject the null

This does not mean we prove the null hypothesis, only that there is **not enough evidence to support alternatives**.

How do we decide to reject the null?

- Statistical tests (e.g. t-test, ANOVA) give us **test statistics**.
- One common example is the **p-value**.
- The p-value is the probability of seeing our experimental data *if the null hypothesis is true*.
- In English: how likely is it that we are seeing this data (result) by chance?
- If this probability is low enough, it means it is unlikely the null hypothesis is true.
- “Low enough” is largely **arbitrary**, typically 5% (so $p=0.05$).

Hypothesis Testing Example

1

Null: "this heart medicine does nothing."

Alternative: "this medicine helps."

2

Do a randomized drug trial giving some patients placebo, some the medicine.

Measure heart rates of all groups.

3

Use a statistical test to determine if the difference between groups is significant.

The test outputs a p-value.

4

If $p < 0.05$, we reject the null hypothesis.

If $p \geq 0.05$, we fail to reject the null hypothesis.

One Versus Two-Tailed Tests

- A one-tailed test means there is a **direction** in the alternative hypothesis.
 - An example is “drug A is better than drug B at managing heart disease”
- A two-tailed test means we are looking for **any** difference, without direction.
 - An example is “There is a difference in heart function between patients who take the heart drug and those who take a placebo” (we’re *specifically not saying* that the drug increases or decreases heart function).

Soda Experiment: Hypothesis Testing

In groups, discuss the following. In the soda experiment:

- ☐ What is the **null** hypothesis?
- ☐ What is the **alternative** hypothesis?
- ☐ Is a one or a two-tailed test appropriate?

Let's see it for ourselves!

Let's open **01-hypothesis-testing-in-action.ipynb**



A/B testing as hypothesis testing

- ❑ A/B tests are **applied hypothesis tests**.
- ❑ Usually used in marketing, web analytics, product development.
- ❑ They usually compare two versions of something (product, website, advert, etc.).
- ❑ The null hypothesis is usually “there is no difference between these versions”.
- ❑ The alternative hypothesis is either:
 - “There is a difference” (two-tailed).
 - Version A is better than B (one-tailed).

Designing an A/B Experiment

Steps of experimental design	A/B test example
1. Define the objectives .	“Do red call to action buttons result in more clicks than the current blue ones?”
2. Identify sources of variation : - Experimental units - Treatment factors - Blocking factors	• Experimental units are individual participants. • The treatment factor is the color of the button. • Blocking factors include demographics (age, gender, etc.), time of day, user device (mobile vs web).
3. Choose a rule to assign units to treatments.	• Participants randomly allocated. Some see the red button, some see the blue one. • Design likely to be between-subject (we don't show the same person different buttons).
4. Decide on measurements .	• Measurement will be % click-through rate. • The null hypothesis is: “There is no difference between colors.” • The alternative hypothesis is: “Red results in more clicks than blue.” ○ We therefore use a one-tailed test.

Designing a Whole Experiment

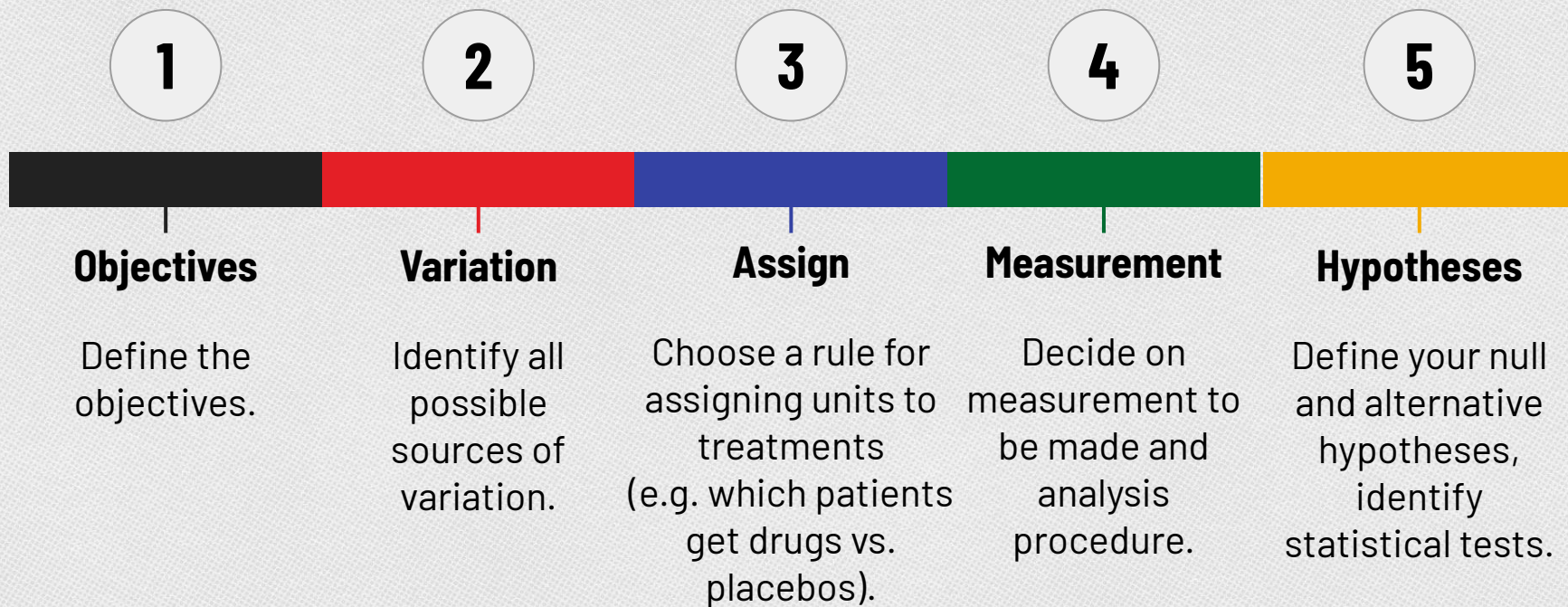
You are working for a sportswear company. They want to know if their **homepage should feature a soccer player or a baseball player** to generate more orders through the website.

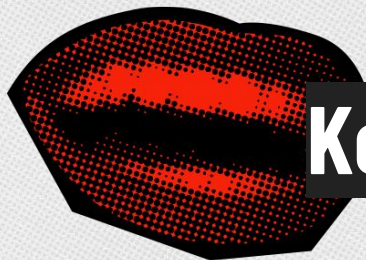
Design the A/B test experiment considering all of the following:

- ☐ The objectives and specific research question(s).
- ☐ Treatment & sources of variation.
- ☐ How treatment groups are assigned.
- ☐ The specific measurements you need to take.
- ☐ The study design (within-subject or between-subject).
- ☐ The null and alternative hypotheses.

Bring It Home

Summary: How to design an experiment





Key takeaways

Designing Experiments

1

Designing Experiments

Remember to consider all aspects.

Think about treatments and factors.

Identify potential sources of bias.

2

Sources of Bias

Bias could arise anywhere from objective setting to factor selection to group assignment!

3

Hypothesis Testing

Remember, we aren't **proving** anything.

We are comparing a **null** (base) hypothesis to alternatives.

Specific statistical methods exist for different problem types.

4

Why Experimental Design is important

Spending time designing an experiment up front sets you up for success!

